Peer Review for

Integrity Trends Analysis Tool (ITAT)

Completed by Statistics Canada

Introduction and Response by IRCC

Immigration, Refugees and Citizenship Canada
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Table of Contents

Introduction to Integrity Trends Analysis Tool by IRCC ……3

Background ……3

ITAT: Purpose, Functionality, Launch ……3

Preface: Insights from the ITAT Peer Review ……..4

Peer Review Report by Statistics Canada ……5

Response for Key Recommendations by IRCC ……7

Introduction to Integrity Trends Analysis Tool by IRCC

Background

Automated tools are part of Immigration, Refugees, and Citizenship Canada (IRCC)'s commitment to using technology responsibly to build a stronger immigration system for all of our clients. In January 2023, IRCC launched a technical system, the Integrity Trends Analysis Tool (ITAT), previously known internally as "Watchtower" or "Lighthouse." ITAT is a data mining tool that leverages advanced analytics techniques to support efficient and effective processing by surfacing risk patterns for investigation by Risk Assessment Units[1]. ITAT enhances the effectiveness and consistency of IRCC's fraud detection and verification activities.

As required by the Treasury Board Secretariat's (TBS) *Directive on Automated Decision-Making*, a peer review of this tool was conducted by the Data Science Team at Statistics Canada.

The goal of the peer review was to provide an informed assessment of whether the technology is being used responsibly by validating IRCC's *Algorithmic Impact Assessment*, the quality of the system, the appropriateness of the quality assurance and risk mitigation measures, as well as to identify the residual risk of operating the system.

IRCC used the findings from this peer review to improve the tool, the tool's governance, and the tool's Algorithmic Impact Assessment (AIA) prior to ITAT's launch in January 2023.

ITAT: Purpose, Functionality, Launch

ITAT analyzes large volumes of data and extracts various risk and fraud patterns within applications containing adverse characteristics, such as inadmissibility findings (e.g. criminality and misrepresentation) and other administrative or enforcement actions related to the Immigration and Refugee Protection Act (IRPA) and *Regulations* (IRPR). Risk Assessment Units are aware that an application is flagged by the tool and decide when to initiate verification activities. However, front line decision-making officers are not informed of ITAT inputs related to the applications that they are processing.

Front line decision-making officers review applications and make decisions consistent with procedural fairness and per the requirements of the IRPA/IRPR. If verification activities are performed, decision-making officers only see verification results and may consider the results when making decisions. Examples of verification activities include authenticating proof of funds from a bank or verifying a letter of acceptance from a Designated Learning Institution (DLI). Consequently, separate and apart from ITAT,

[1] Risk Assessment Unit are groups of specialized officers who focus on evaluating and managing risks related to immigration and refugee applications. They identify fraud trends and verify the authenticity of documents, ensuring that the work of front-line decision-makers is grounded in a comprehensvie understanding of potential risks.

a processing officer may request a verification activity, or it may be initiated independently by a Risk Assessment Unit if a fraud trend is identified.

The Risk Assessment Units may use various techniques including telephone calls, site visits, or in-person interviews as a means to validate the risks. The results of the verification activities are shared with decision-making officers whether or not adverse information was found. ITAT is designed to assist Risk Assessment Units in identifying applications for verification activity by automating the detection of risk patterns associated with adverse outcomes from historical data and flagging new applications that match these patterns.

ITAT's risk patterns are not visible to decision-making officers. Rather, decision-making officers only receive the results of the verification activities performed by Risk Assessment Units, which may have been initiated with or without the tool. As such, ITAT does not automate recommendations or decisions on applications.

In 2023, ITAT for the temporary resident programs was implemented in a phased approach. It was initially used on study permit applications and subsequently expanded to temporary resident visa applications and temporary foreign worker applications. The subsequent phase will involve the integration of eTA (electronic travel authorization). Continuous evaluation will be undertaken to ensure ongoing assurance of the tool's performance and responsible use. This approach seems systematic and thorough, ensuring the tool's effectiveness while also maintaining accountability.

Preface: Insights from the ITAT Peer Review

IRCC used the findings from this peer review, completed in 2021, to improve the tool, the tool's governance, and the tool's Algorithmic Impact Assessment (AIA) prior to ITAT's launch in January 2023.

As the basis for the peer review, Statistics Canada's Data Science Division reviewed many aspects of the tool and its governance, including ITAT's *Algorithmic Impact Assessment* and the proposed quality assurance plan for monitoring the tool post-deployment. Statistics Canada's Data Science Division also evaluated documents outlining the purpose of ITAT, the business requirements it is meant to satisfy, and the algorithms governing its operation. These documents provided a detailed understanding of the data ITAT takes as input, the logical rules the tool applies to this data, and the resulting outputs.

Peer Review Report by Statistics Canada

Introduction

This document provides a peer review of the Watchtower Risk Identification Prototype. The core purpose is to inform the team responsible for this prototype of potential ethical or quality concerns. The review has been done by relying on Statistics Canada (2021) Responsible Machine Learning Framework. Bronson and Millar (2020) propose a high-level process for peer review under Canada's Directive on Automated Decision-Making which was also consulted. Moreover, we have examined the Algorithmic Impact Assessment form as it has been filled by the team responsible for the Watchtower project.

Key Recommendations:

1. We recommend describing, with concrete examples, the kind of decisions/actions that the IRCC officials will be able to make/take with this tool in order for the readers to better understand the benefits and risks associated with this project. What will the agents do with the information? What can happen to the applicants? Are the applicants in Canada or a different country? It would be good to define the expression 'adverse information' more rigorously. Is it an euphemism for 'crime' or 'offense'?

2. The algorithm associates "data patterns to adverse information" p.5. We recommend using more standard terminology such as "independent variables to dependent variables" or "features to labels". This will improve the interpretability of the documentation of the project.

3. Based on the information in the document, this looks like a predictive machine learning (ML) project where one models the relationships between the variables in the dataset. How can this tool provide "fact-based information that end users (IRCC officers) may find relevant for their adjudication process" p.5 if the officers are not going to classify/make predictions on individuals? Expressions such as "identify only those patterns suggestive of organized risk" p.8 imply predictive labeling. Therefore, we recommend removing expressions such as "the tool is not predictive"p.5.

4. If this is a predictive modelling project, then a train/validation/test methodology must be applied. If there is no need for validation, then it should be explained. The metrics that will be used should be described and justified. How do you measure the "utility and timeliness" p.14?

5. We recommend a more thorough description of the quality of the data set that will be used by the algorithm. Does the data fail to cover a subpopulation that could be at risk? Are there any missing data? If so, has there been any kind of imputation (without leakage)? Do you use all of the available data to run the algorithm or only the observations that have been previously flagged by officers?

6. We recommend answering the following two questions. Will the information used reflect and exacerbate stereotypes? We understand that the purpose is not to identify a "broad subpopulation that tends to have higher adverse rates" but will someone monitor this over time?

7. We recommend describing how the quality of the algorithm will be maintained over time. How will you measure the “utility and timeliness” p.14 over time? Will an individual be investigated over and over

8. It would be good to describe the alternative methods that have been considered, like decision trees, and to explain why they have not been chosen.

9. We recommend that this report should be approved by the Deputy Minister in charge of the program that will use the automated decision system.

10. We recommend the publication of this report and the completed Algorithmic Impact Assessment on the Open Government Portal prior to deploying the system.

Minor Recommendations:

1. We recommend explaining how this tool will ultimately benefit Canadians in terms that they will understand. This can help motivate the need for such measures.
2. We recommend indicating that some research has been done (bibliography) on the ethical use of machine learning for decision-making with crime data**.**

Response for Key Recommendations by IRCC

StatsCan Key Recommendations	IRCC Response for Key Recommendations:
1. We recommend describing, with concrete examples, the kind of decisions/actions that the IRCC officials will be able to make/take with this tool in order for the readers to better understand the benefits and risks associated with this project. What will the agents do with the information? What can happen to the applicants? Are the applicants in Canada or a different country? It would be good to define the expression 'adverse information' more rigorously. Is it an euphemism for 'crime' or 'offense'?	**1.** IRCC has developed a few descriptive use-case stories/scenarios and a process-flow diagram to clarify the benefits and the associated risks with the intended pilot use-case for the tool. Additionally, we included further documentation specifying the types of adverse information utilized by the system. During the pilot, the tool provided Risk Assessment officers (RAO) with information that may aid in the decision-making or when to seek further verifications regarding an application, i.e. RAO contacts a bank or school to validate the authenticity of a document provided by an applicant. This decision about when to undertake this verification rests with the RAO; the tool summarizes information from IRCC's database to help inform that decision.
2. The algorithm associates "data patterns to adverse information" p.5. We recommend using more standard terminology such as "independent variables to dependent variables" or "features to labels." This will improve the interpretability of the documentation of the project.	**2.** The approach was taken to ensure relevance to a general audience; however, IRCC made the appropriate modifications and used more standard terminology.
3. Based on the information in the document, this looks like a predictive machine learning (ML) project where one models the relationships between the variables in the dataset. How can this tool provide "fact-based information that end users (IRCC officers) may find relevant for their adjudication process" p.5 if the officers are not going to classify/make predictions on individuals? Expressions such as "identify only those patterns suggestive of organized risk" p.8 imply predictive labeling. Therefore, we recommend removing expressions such as "the tool is not predictive"p.5.	**3.** IRCC adjusted the wording to clarify the predictive nature of the system. To clarify, the tool does not make recommendations and is not used directly to influence administrative decision-making on applications. It does not generate recommendations or provide predictive scores. Instead, it summarizes factual information from IRCC's data holdings on applications where a possible concern is identified. The predictive aspect lies in the fact that information is shown only for applications that match previously identified risk patterns, based on the data. While the system identifies data from IRCC's data holding, it is essential to note that the factual information provided should stand on its own merits and be evaluated independently. Decision-making officers rely solely on factual evidence provided to them by RAOs, following verification activity, never on ITAT algorithms or risk patterns. This distinction aims to differentiate the approach from traditional predictive systems. For added context, it is important to note that the existing process for verifying application information is roughly the same as presented above, except the decisions about when to conduct such document verifications are made in a more *ad hoc* manner, relying on officer experience and not

	leveraging evidence from IRCC's data holdings. ITAT is intended to improve this existing business process by putting additional fact-based information into the hands of RAO officers while performing these verifications.
4. If this is a predictive modeling project, then a train/validation/test methodology must be applied. If there is no need for validation, then it should be explained. The metrics that will be used should be described and justified. How do you measure the "utility and timeliness" p.14?	**4.** There are two main reasons why the traditional validation methodologies are not particularly valuable for assessing the "utility and timeliness" of this predictive modeling project: a) Lack of ground truth labels: Due to a low sample size and rate of verifications, currently performed on only 4% of the caseload, which means labeled data is scarce for retrospective analysis. The insufficient instances where the true outcomes (ground truth labels) are known make it unreliable for simulating operational performance. b) Preference for qualitative user testing over traditional validation methods: This approach involves close collaboration with end users to manually assess historical patterns identified by the system. The system generates interpretable historical pattern information, enabling risk assessment officers to make assessments solely on the face of the information presented. This contrasts against typical ML systems that would use less interpretable output (binary recommendations, scores, etc.) where there is a much higher reliance on showing test-sample performance as a means to validate the system. This qualitative user-testing approach is in many ways superior to using quantitative test set results, particularly given the lack of ground truth labels. It also reinforces the point that the system should merely identify factual information that can stand on its own merits, rather than expecting reliance on past out-of-sample test performance to validate the usage. It emphasizes presenting relevant information rather than predicting outcomes, providing a more relevant performance assessment.
5. We recommend a more thorough description of the quality of the data set that will be used by the algorithm. Does the data fail to cover a subpopulation that could be at risk? Are there any missing data? If so, has there been any kind of imputation (without leakage)? Do you use all of the available data to run the algorithm or only the observations that have been previously flagged by officers?	**5.** IRCC prioritized qualitative user testing for validation, enhanced collaboration with end-users, and maintained transparency about data quality and use while committing to no data imputation. All data is used in the algorithm. Depending on the specification of the target variable for a particular project, there are some relevant nuances to how the data itself is generated. For example, decisions on when to perform a verification (i.e., seek additional information from a client or to validate the authenticity of documents they submitted) are somewhat region-specific, and while some randomly chosen verifications are occasionally performed, we know that the current verification selection approach will impact the tool's outputs.
6. We recommend answering the following two questions. Will the information used reflect and exacerbate stereotypes? We understand that the purpose is not to identify a "broad subpopulation that tends to have higher	**6.** Our approach to address all concerns of perpetuating stereotypes and biases is primarily by examining false positive rates across demographic variables. To mitigate stereotypes, risk assessment officers will only use evidence from neutral, non-subjective third-party sources. For example, an illustrative scenario would involve a Risk Assessment Officer receiving a summary of historical adverse patterns related to a student permit application; the officer then determines whether to independently verify the applicant's

adverse rates" but will someone monitor this over time?	enrollment status with the alleged school. Then, it is the front line decision-making officer who sees the results of the verification process without any awareness that ITAT may have flagged the application as matching a historical risk pattern and makes the decision solely based on evidence specific to the current applicant therefore minimizing bias or the perpetuation of stereotypes. More importantly, the impact of a false positive on the system is minimal and possibly even favorable for the applicant. In the above example, a false positive would involve the system identifying a pattern that, upon human review, is deemed sufficient to merit conducting a verification with the school and with that verification confirming the applicant is indeed registered. In this scenario, the impact is possibly a slight delay in processing but increasing the applicant's likelihood of approval. This careful design aims to mitigate any negative consequences from false positives.
7. We recommend describing how the quality of the algorithm will be maintained over time. How will you measure the "utility and timeliness" p.14 over time? Will an individual be investigated over and over?	**7.** The high-level plan is to regularly shift the time window in which patterns are assessed, as well as to qualitatively assess the utility of the tool for risk assessment officers. This ensures that the system is always displaying relevant and timely data and that stale information is not being presented. Ongoing mitigations will be required to ensure that the system does not self-reinforce its past results. We have had discussions about the possibility of using random sampling methods for verifications to support future training data, but there are operational/resource considerations around doing so.
8. It would be good to describe the alternative methods that have been considered, like decision trees, and to explain why they have not been chosen.	**8.** Early testing considered several alternative methods, including C5 decision trees. However, our current approach, association rule mining, is preferred for two primary reasons related to how patterns are identified and the need for transparency. The first reason pertains to how decision trees and association rules process data. Decision trees partition the dataset by recursively splitting it into mutually exclusive nodes, aiming to maximize predictive accuracy or minimize classification error. The structure limits their ability to identify smaller, highly associated clusters of applications. In contrast, association rule mining focuses on detecting strong relationships within localized subsets of data, without the restriction of mutual exclusivity. This makes it particularly effective at uncovering small but meaningful patterns that may not exhibit broad, dataset-wide connections. The second reason is the need for transparency and interpretability. Concerns were raised about the lack of transparency in decision trees and other algorithmic options, making it difficult to clearly understand or explain the patterns they produce. Since full transparency is essential for this system, association rule mining is the preferred approach.

	Other methods were also considered but were found to be less effective at capturing the complex, interrelated patterns required for the analysis.
9. We recommend that this report should be approved by the Deputy Minister in charge of the program that will use the automated decision system.	**9.** In accordance with the TBS Directive on Automated Decision-Making the Assistant Deputy Minister (ADM) is responsible for the Algorithmic Impact Assessment and applicable requirements (i.e., the peer review.) To transition from pilot to permanent tool, the Senior ADM of Service Delivery and SADM of Strategic and Digital Policy approved a memorandum in September 2022. This memo approved publication of the AIA to the Open Government Portal and reviewed all TBS and IRCC governance requirements. Approval to launch was given by the Director General Steering Committee. This peer review was circulated and responded to throughout the governance process but Deputy Minister approval as a distinct document was not required by TBS. That being said, in accordance with this recommendation, IRCC will share the peer review report with our deputy minister for approval. In January 2023 the tool was phased into the temporary resident programs: students, followed by visitors, and concluding with workers. Continuous evaluation and reporting to senior management at IRCC will be undertaken to ensure ongoing assurance of the tool's performance and responsible utilization.
10. We recommend the publication of this report and the completed Algorithmic Impact Assessment on the Open Government Portal prior to deploying the system.	**10.** This report will be published with the completed Algorithmic Impact Assessment on the Open Government Portal
StatsCan Minor Recommendations	**IRCC Response for Minor Recommendations:**
1. We recommend explaining how this tool will ultimately benefit Canadians in terms that they will understand. This can help motivate the need for such measures.	**1.** Indeed, the revised AIA addresses this recommendation and will explain how this tool ultimately benefits Canadians through improved program integrity and efficient application process.
2. We recommend indicating that some research has been done (bibliography) on the ethical use of machine learning for decision-making with crime data**.**	**2.** Please see response #6 above about the impact on clients of false positives. We believe that some policing examples may not be directly comparable because they often carry significant negative impacts in the event of false positives. We will examine the research and draw out some of the comparisons and contrasts.